# Benford's Law and Applications

Math Capstone

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# Benford's Law

There are many different applications of Benford's Law, but this article will more specifically analyze the application of Benford's Law in large data sets of all kinds. Such as tax return data and population.

## Introduction:

First-digit law, or better known as Benford's Law, is a law revolving around the distribution of the first digits in data. Benford's Law is a mathematical formula that is used to determine the frequency of digits in a given data set. Mostly used in large data sets, Benford's Law can be used in anything from population distributions to tax fraud as it is more commonly used today.

In this paper it is my goal to help the public better understand the application and foundation of one of the most prevalent laws involving countable numbers. While this law is known, it is a very common misconception to assume that the distribution of numbers in a data set is truly random. Benford's Law explains the distribution of those numbers and proves that over a vast data set, the distribution is not random.

# Background:

In the article "The Mathematics of Benford's Law – A Primer" by Arno Berger and Theodore P. Hill, they go in depth about the foundational mathematics and notation used to create the formula. Not all data can be represented by the first-digit law, and there are very specific circumstances where Benford's Law is applicable. The Proposition 7 as stated in "The Mathematics of Benford's Law – A Primer" is the best example of the formula for Benford's Law.

**Proposition 7.** *A sequence ($x_n$) of real numbers is Benford if and only if*

$$\lim_{N\to\infty}\left(\frac{\#\{1 \le n \le N :\ D_1(x_n) = d_1, D_2(x_n) = d_2, \dots, D_m(x_n) = d_m\}}{N}\right)$$

$$= \log\left(1 + \frac{1}{10^{m-1}d_1 + 10^{m-2}d_2 + \cdots + d_m}\right)$$

*for all* $m \in N$, *all* $d_1 \in \{1, 2, \dots, 9\}$, *and all* $d_j \in \{0, 1, \dots, 9\}, j \ge 2.$

Proposition 7 with m = 1 yields the well-known *first-digit law*: For every Benford sequence of real numbers ($x_n$),

$$\lim_{N\to\infty}\left(\frac{\#\{1 \le n \le N :\ D_1(x_n) = d\}}{N}\right)$$

$$= \log(1 + \tfrac{1}{d}) \text{ for all } d \in \{1, 2, \dots, 9\}.$$

This formula gives the approximation for the distribution of leading digits in data sets. The variable "d" is used as the first digit in a given piece of data, i.e., for 19,242 d = 1.

## Definitions and Theorems:

In this section I will be stating the major theorems and definitions behind the statistical formula of Benford's Law. A lot of the applications in the formulas are things taught in Intermediate Statistic classes as well as Calculus classes.

**Definition 1:** The first (decimal) significant digit of x ∈ R, denoted $D_1$(x), is the first (left most) digit of S(x), where by convention the terminating decimal representation is used if S(x) 2 has two decimal representations. Similarly, $D_2$(x) denotes the second digit of S(x), $D_3$(x) the third digit of S(x), and so on. (Note that $D_n(0) = 0$ for all n ∈ N.)

**Definition 2:** A sequence of real numbers $(x_n)$ is a Benford sequence, or Benford for short, if for every t ∈ [1, 10), the limiting proportion of $x_n$'s with significand less than or equal to t is exactly log t, i.e., if $\lim_{N\to\infty} \frac{\#\{1 \le n \le N : S(x_n) \le t\}}{N} = \log t \quad \text{for all } t \in [1, 10).$

In "The Mathematics of Benford's Law – A Primer" by Arno Berger and Theodore P. Hill, they stated multiple limiting properties of sequences of random variables. "These include the three basic facts that:

(i) powers of every continuous random variable converge to Benford's law;

(ii) products of random samples from every continuous distribution converge to Benford's law;

(iii) if random samples are taken from random distributions that are chosen in an unbiased way, then the combined sample converges to Benford's law. (Berger and Hill, 11).

All sequences of random variables will have these three properties, if a given sequence of random variables does not follow these three properties then it is not applicable for Benford's Law.

**The Law of Large Numbers:** The law of large numbers theorem states that if the same experiment or study is repeated independently a large number of times, the average of the results of the trials must be close to the expected value. The result becomes closer to the expected value as the number of trials increases. Which is evident as Benford's Law becomes more applicable as the data sets grow larger.

**Frequency of Digits in the qth Place:** The frequency of digits in the qth place is best explained by Frank Benford in his journal about the Law of Anomalous Numbers, "The second-place digits are ten in number, for here we must take 0 into account. Also, in considering the frequency $F_b$ of a second-place digit b we must take into account the digit a that preceded it. The logarithmic interval between two digits is now to be divided into ten parts corresponding to the ten digits 0, 1, 2, ... 9. Let a be the first digit of a number and b be the second digit; then using the customary meaning of position and order in our decimal system a two-digit number is written ab, and the next greater number is written ab + 1. The logarithmic interval between ab and ab + 1 is log (ab + 1) - log ab, while the interval covered by the ten possible second-place digits is log (a + 1) - log a. Therefore the frequency $F_b$ of a second-place digit b following a first-place digit a is

$$F_b = \log\left(\frac{ab+1}{ab}\right) \Big/ \log\frac{a+1}{a}.$$

As an example, the probability $F_b$ of a 0 following a first-place 5 in a random number is the quotient

$$F_b = \log\frac{51}{50} \Big/ \log\frac{6}{5}.$$

It follows that the probability for a digit in the qth position is

$$F_b = \frac{\log\dfrac{abc\cdots p\,(q+1)}{abc\cdots pq}}{\log\dfrac{abc\cdots o\,(p+1)}{abc\cdots p}}.$$

Here the frequency of q depends upon all the digits that precede it, but when all possible combinations of these digits are taken into account $F_q$ approaches equality for all the digits 0, 1, 2, *.. 9, or $F_q \doteq 0.1$. As a result of this approach to uniformity in the qth place the distribution of digits in all places in an extensive tabula- tion of multi-digit numbers will be also nearly uniform." (Benford, 555).

**The Law of Anomalous Numbers:** Frank Benford discuses the overarching theorem around the first-digit law by explaining how the Law of Anomalous numbers functions. "A study of the items of Table I shows a distinct tendency for those of a random nature to agree better with the logarithmic law than those of a formal or mathematical nature. The best agreement was found in the arabic numbers (not spelled out) of consecutive front page news items of a newspaper. Dates were barred as not being variable, and the omission of spelled-out numbers restricted the counted digits to numbers 10 and over. The first 342 street addresses given in the cur- rent American Men of Science (Item R, Table IV) gave excel- lent agreement, and a complete count (except for dates and page numbers) of an issue of the Readers' Digest was also in agreement. On the other hand, the greatest variations from the logarithmic relation were found in the first digits of mathematical tables from engineering handbooks, and in tabulations of such closely knit data as Molecular Weights, Specific Heats, Physical Constants and Atomic Weights.

TABLE IV

SUMMATION OF DIFFERENCES BETWEEN OBSERVED AND THEORETICAL FREQUENCIES

| Order | Item | Nature | Difference | Order | Item | Nature | Difference |
|---|---|---|---|---|---|---|---|
| 1 | D | Newspaper Items | 2.8 | 11 | N | Cost Data, Concrete | 12.4 |
| 2 | F | Pressure Lost, Air Flow | 3.2 | 12 | S | $n^1 \cdots n^8$, $n!$ | 13.8 |
| 3 | G | H.P. Lost in Air Flow | 4.8 | 13 | L | Design Data Generators | 16.6 |
| 4 | R | Street Addresses, A.M.S. | 5.4 | 14 | B | Population, U. S. A. | 16.6 |
| 5 | P | Am. League, 1936 | 6.6 | 15 | I | Drainage Rate of Rivers | 21.6 |
| 6 | Q | Black Body Radiation | 7.2 | 16 | K | $n^{-1}$, $\sqrt{n} \cdots$ | 22.8 |
| 7 | O | X-Ray Voltage | 7.4 | 17 | H | Molecular Wgts. | 23.2 |
| 8 | M | *Readers' Digest* | 8.4 | 18 | E | Specific Heats | 24.2 |
| 9 | A | Area Rivers | 9.8 | 19 | C | Physical Constants | 34.9 |
| 10 | T | Death Rates | 11.2 | 20 | J | Atomic Weights | 35.4 |

These facts lead to the conclusion that the logarithmic law applies particularly to those outlaw numbers that are without known relationship rather than to those that individually follow an orderly course; and therefore the logarithmic relation is essentially a Law of Anomalous Numbers." (Benford, 556-557). This theorem explains how Benford's Law is just an extension of the Law of Anomalous Numbers.

# Applications:

Like most statistical laws, they are only representatives of vast data sets. If there is a set of data that has too small of a sample size, it may not be fully representative of the law, thus the law would not be applicable in that instance. Benford's Law is mostly used to detect tax fraud. The way that Benford's Law is able to detect tax fraud is by analyzing the first digit of all of the amounts of money in a given account. People committing tax fraud will likely fabricate their tax returns in order to keep/make more money from taxes. With fabricated returns, there are a lot of discrepancies in the actual tax return. Benford's Law is applied to analyze all of the different amounts in a given tax year, and if the variation of numbers differs too heavily from the expected value, these tax returns can be flagged for higher inspection by tax agencies. Using the formula for Benford's Law, plugging in each digit for "d" in

$\log(1 + \frac{1}{d})$ for all $d \in \{1, 2, \ldots, 9\}$, you will get the following percentages shown below.

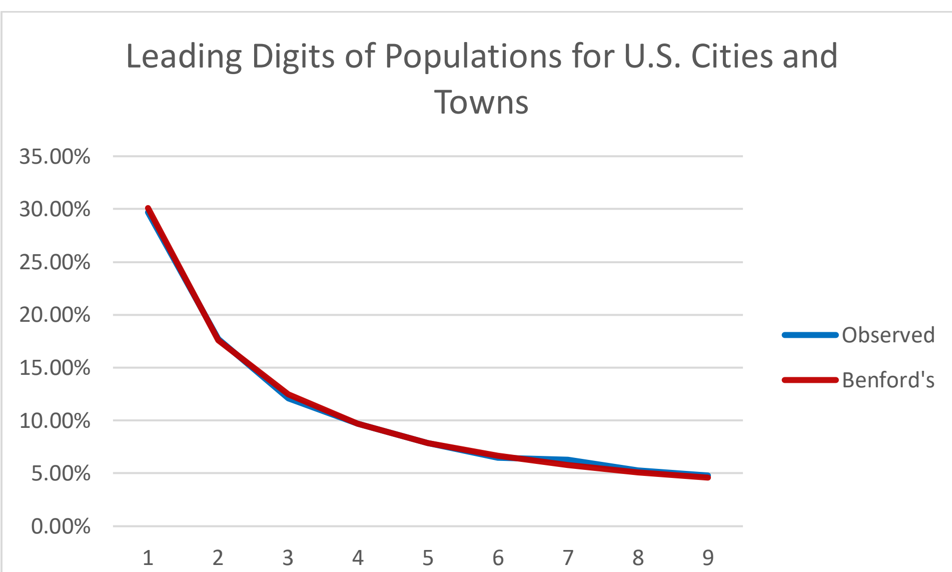
| Digits | Benford's |
|---|---|
| 1 | 30.10% |
| 2 | 17.60% |
| 3 | 12.50% |
| 4 | 9.70% |
| 5 | 7.90% |
| 6 | 6.70% |
| 7 | 5.80% |
| 8 | 5.10% |
| 9 | 4.60% |

These percentages are to represent how the numbers should fall in a sequence of random variables. Statisticians use these percentages and compare it to the percentages observed in various data sets such as tax returns.

I wanted to test how accurate Benford's Law is and see if it can be applied to things such as populations. Instead of just picking a handful of places, I chose to use the population of every city and town in the United States. Which totals to 19,486 data points of population. I ran my own test by writing the leading digit of all of the populations, and then calculating the observed percentages of each digit. I came to find this data:

| Digits | Count | Observed | Benford's |
|---|---|---|---|
| 1 | 5786 | 29.70% | 30.10% |
| 2 | 3450 | 17.70% | 17.60% |
| 3 | 2350 | 12.10% | 12.50% |
| 4 | 1895 | 9.70% | 9.70% |
| 5 | 1530 | 7.90% | 7.90% |
| 6 | 1272 | 6.50% | 6.70% |
| 7 | 1229 | 6.30% | 5.80% |
| 8 | 1041 | 5.30% | 5.10% |
| 9 | 933 | 4.80% | 4.60% |

In comparison to Benford's Law, the observed percentages of the populations of all cities and towns in the United States is actually extremely close to the expected value given by Benford's Law.



This line graph compares the observed value against Benford's, and as we can see the lines are almost identical. This leads me to believe Benford's Law can also be used to predict the distribution of populations and all things population related. By the Law of Large Numbers, since the sample size of the chosen data of the population of all United States towns and cities, the data should correlate and slowly approach that of Benford's Law if the data supports the law. Which in this case the data approaches the expected values of Benford's Law; so we can say that populations can also be represented through Benford's Law.

The way this relates to classes taken is that the formula itself is a log based formula which are mostly introduced and taught in calculus. The applications of Benford's Law are purely statistical since Benford's Law deals with probabilities for each digit, so it has a very high statistical component. Throughout my research into the utilization of Benford's Law, I found that it is applicable in things from Fraud detection to things as simple as likes on YouTube. If the sample size is large enough, most if not all data points will follow the law. Any sequence of random variables that can be represented numerically, will have a distribution similar to that of the first-digit law. A few pieces of my data from the population give a rough estimate of how long it takes to count it all by hand.

| | |
|---|---|
| Abbeville city, Alabama | 2,368 |
| Adamsville city, Alabama | 4,372 |
| Addison town, Alabama | 664 |
| Akron town, Alabama | 227 |
| Alabaster city, Alabama | 33,360 |
| Albertville city, Alabama | 22,390 |
| Alexander City city, Alabama | 14,807 |
| Aliceville city, Alabama | 2,175 |
| Allgood town, Alabama | 544 |
| Altoona town, Alabama | 954 |
| Andalusia city, Alabama | 8,739 |

The best way to count all of the coefficients is using software given in excel. It takes a bit of knowledge of Microsoft Excel, but there are functions that will isolate only the first digit of a given set of data, and from there input another program that will count every one, two, three, … , all the way to nine. The percentage calculation must be done by hand, but it is a simple division by the total number of data points. Here I will post a link to all 19,486 data points with their corresponding values. Populations of all U.S. Towns and Cities.xlsx

**Conclusion:** Benford's Law is applicable in many instances and it gives a very interesting insight into how the world works. Normally people would assume things such as the amount of money they get paid and the amount that they withdraw/deposit is completely random. Benford's

Law says that it is not random and instead it follows a very specific formula. This formula helps maintain order and security for all things. As of now it is used mainly for Fraud Detection, but in the future, I believe it could be used for even greater things such as medical records, prescriptions, dosages, and things like that, which would help the world greatly. All of these things will take a lot more time and research to discover, but since the Law is naturally occurring it is very possible.

# References:

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